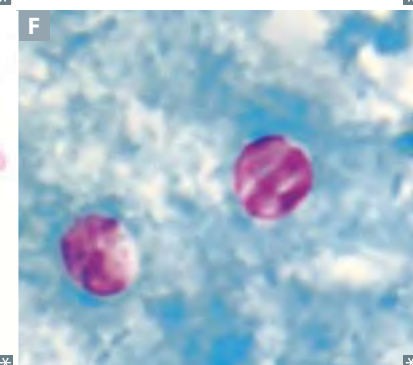
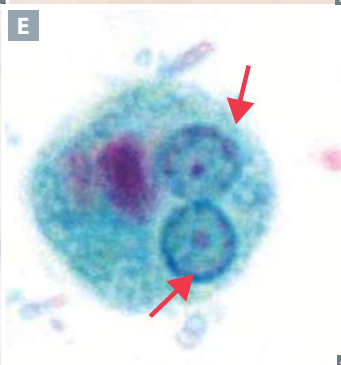
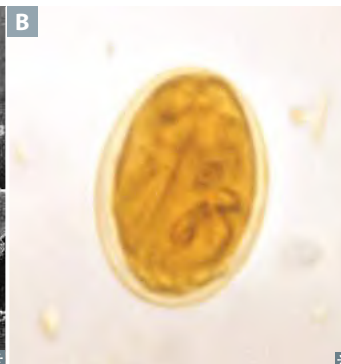
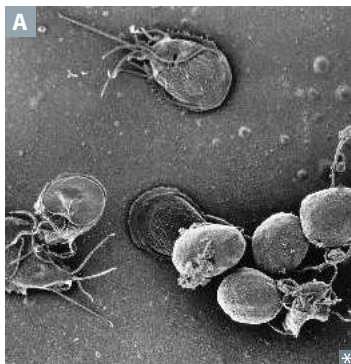


► MICROBIOLOGY—PARASITOLOGY

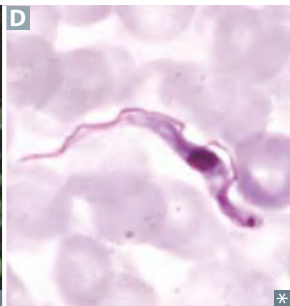
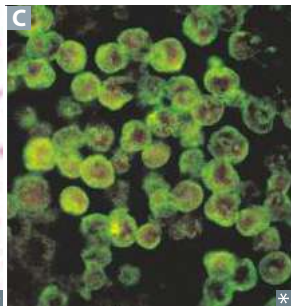
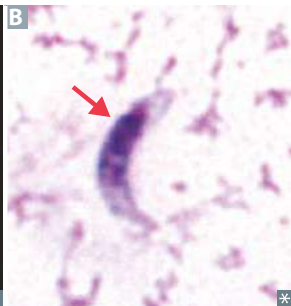
Protozoa—gastrointestinal infections

ORGANISM	DISEASE	TRANSMISSION	DIAGNOSIS	TREATMENT
<i>Giardia lamblia</i>	Giardiasis —bloating, flatulence, foul-smelling, nonbloody, fatty diarrhea (often seen in campers/hikers)—think fat -rich Chirardelli chocolates for fatty stools of Giardia	Cysts in water	Multinucleated trophozoites A or cysts B in stool, antigen detection, PCR	Tinidazole, nitazoxanide, or metronidazole
<i>Entamoeba histolytica</i>	Amebiasis —bloody diarrhea (dysentery), liver abscess (“anchovy paste” exudate), RUQ pain; histology of colon biopsy shows flask-shaped ulcers C	Cysts in water	Serology, antigen testing, PCR, and/or trophozoites (with engulfed RBCs D in the cytoplasm) or cysts with up to 4 nuclei in stool E ; Entamoeba Eats Erythrocytes	Metronidazole; paromomycin for asymptomatic cyst passers
<i>Cryptosporidium</i>	Severe diarrhea in AIDS Mild disease (watery diarrhea) in immunocompetent hosts	Oocysts in water	Oocysts on acid-fast stain F , antigen detection, PCR	Prevention (eg, filtering); nitazoxanide (severe disease and/or immunocompromised)



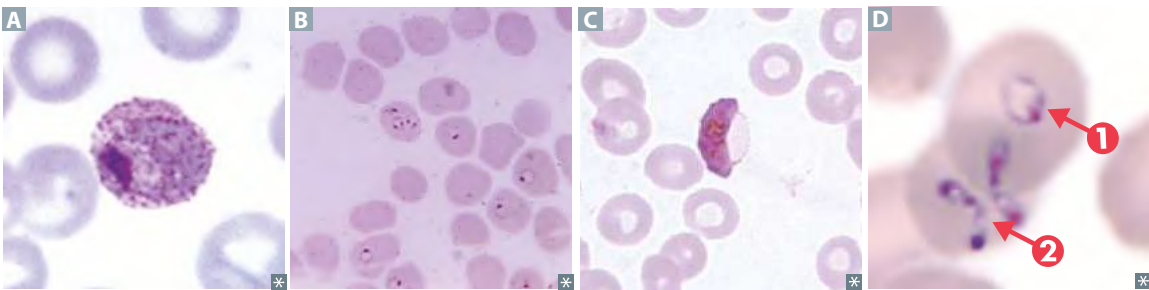
Protozoa—CNS infections

ORGANISM	DISEASE	TRANSMISSION	DIAGNOSIS	TREATMENT
<i>Toxoplasma gondii</i>	Immunocompetent: mononucleosis-like symptoms, ⊖ heterophile antibody test Reactivation in AIDS → brain abscesses usually seen as multiple ring-enhancing lesions on MRI A Congenital toxoplasmosis: classic triad of chorioretinitis, hydrocephalus, and intracranial calcifications	Cysts in meat (most common); oocysts in cat feces; crosses placenta (pregnant patients should avoid cats)	Serology, biopsy (tachyzoite) B ; PCR of amniotic fluid for possible intrauterine disease	Sulfadiazine + pyrimethamine Prophylaxis with TMP-SMX when CD4+ cell count < 100 cells/mm ³
<i>Naegleria fowleri</i>	Rapidly fatal meningoencephalitis	Swimming in warm freshwater; enters CNS through olfactory nerve via cribriform plate	Amoebas in CSF C	Amphotericin B has been effective for a few survivors
<i>Trypanosoma brucei</i>	African sleeping sickness — enlarged lymph nodes, recurring fever (due to antigenic variation), somnolence, coma	Tsetse fly, a painful bite	Trypomastigote in blood smear D	Suramin for blood- borne disease or melarsoprol for CNS penetration ("I sure am mellow when I'm sleeping ")



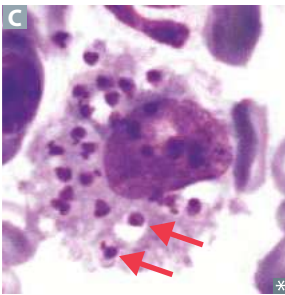
Protozoa—hematologic infections

ORGANISM	DISEASE	TRANSMISSION	DIAGNOSIS	TREATMENT
<i>Plasmodium</i>	Malaria —cyclic fevers, headache, anemia, splenomegaly; hypoglycemia in severe disease	<i>Anopheles</i> mosquito	Peripheral blood smear (also allows for identification of species)	If sensitive, chloroquine; if resistant, mefloquine, doxycycline or atovaquone/proguanil If life threatening, use intravenous quinine or artesunate (test for G6PD deficiency)
<i>P. malariae</i>	72-hr fever cycle (quartan)		Trophozoite ring within RBC	
<i>P. vivax/ovale</i>	48-hr fever cycle (tertian); dormant form (hypnozoite) in liver		Trophozoites and Schüffner stippling (small red granules) within RBC cytoplasm A	Add primaquine to target hypnozoites
<i>P. falciparum</i>	Severe, irregular fever pattern; parasitized RBCs may occlude capillaries in brain (cerebral malaria), kidneys, lungs		Trophozoite ring (headphone shaped) within RBC B ; crescent-shaped gametocytes C	
<i>Babesia</i>	Babesiosis —fever and hemolytic anemia; predominantly in northeastern and north central United States; asplenia ↑ risk of severe disease due to inability to clear infected RBCs	<i>Ixodes</i> tick (also vector for <i>Borrelia burgdorferi</i> and <i>Anaplasma</i> spp)	Ring form D1 , “Maltese cross” D2 ; PCR	Atovaquone + azithromycin



Protozoa—others

ORGANISM	DISEASE	TRANSMISSION	DIAGNOSIS	TREATMENT
Visceral infections				
<i>Trypanosoma cruzi</i>	Chagas disease— dilated cardiomyopathy with apical atrophy, mega colon, mega esophagus; (<i>T cruzi</i> causes big problems); predominantly in South America Unilateral periorbital swelling (Romaña sign) characteristic of acute stage	Triatomine insect (kissing bug) bites and defecates around the mouth or eyes → fecal transmission into bite site or mucosa	Trypomastigote in blood smear A	Benznidazole or nifurtimox
<i>Leishmania</i> spp	Visceral leishmaniasis (kala-azar)—spiking fevers, hepatosplenomegaly, pancytopenia Cutaneous leishmaniasis—skin ulcers B	Sandfly	Macrophages containing amastigotes C	Amphotericin B, sodium stibogluconate
Sexually transmitted infections				
<i>Trichomonas vaginalis</i>	Vaginitis—foul-smelling, greenish discharge; itching and burning; do not confuse with <i>Gardnerella vaginalis</i> , a gram-variable bacterium associated with bacterial vaginosis	Sexual (cannot exist outside human because it cannot form cysts)	Trophozoites (motile) D on wet mount; punctate cervical hemorrhages (“strawberry cervix”)	Metronidazole for patient and partner(s) (prophylaxis; check for STI)



Nematode routes of infection

Ingested—**E**nterobius, **A**scaris, **T**oxocara, **T**richinella, **T**richuris
Cutaneous—**S**trongyloides, **A**ncylostoma, **N**ecator
Bites—**L**oa loa, **O**nchocerca volvulus, **W**uchereria bancrofti

You’ll get sick if you **EATTT** these!

These get into your feet from the **SANd**

Lay **LOW** to avoid getting bitten

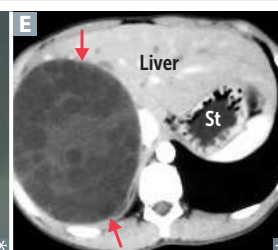
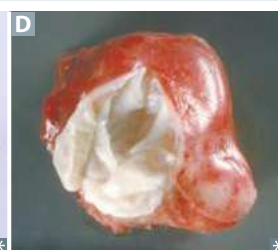
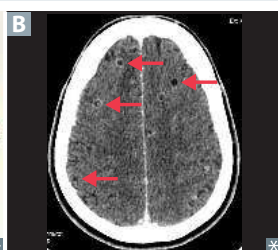
Nematodes (roundworms)

ORGANISM	DISEASE	TRANSMISSION	TREATMENT
Intestinal			
<i>Enterobius vermicularis</i> (pinworm)	Causes anal pruritus, worse at night (eggs A visualized via tape test). Most common in children aged 5–10.	Fecal-oral.	Bendazoles, pyrantel pamoate.
<i>Ascaris lumbricoides</i> (giant roundworm)	May cause obstruction at ileocecal valve, biliary obstruction, intestinal perforation, migrates from nose/mouth. Migration of larvae to alveoli → Löeffler syndrome (pulmonary eosinophilia).	Fecal-oral; knobby-coated, oval eggs seen in feces under microscope B .	Bendazoles.
<i>Strongyloides stercoralis</i> (threadworm)	GI (eg, duodenitis), pulmonary (eg, dry cough, hemoptysis), and cutaneous (eg, pruritus) symptoms. Hyperinfection syndrome can be caused by accelerated autoinfection in the immunocompromised.	Larvae in soil penetrate skin; rhabditiform larvae seen in feces under microscope.	Ivermectin or bendazoles.
<i>Ancylostoma</i> spp, <i>Necator americanus</i> (hookworms)	Cause microcytic anemia by sucking blood from intestinal wall. Cutaneous larva migrans—pruritic, serpiginous rash C .	Larvae penetrate skin from walking barefoot on contaminated beach/soil.	Bendazoles or pyrantel pamoate.
<i>Trichinella spiralis</i>	Larvae enter bloodstream, encyst in striated muscle D → myositis. Trichinosis—fever, vomiting, nausea, periorbital edema, myalgia.	Undercooked meat (especially pork); fecal-oral (less likely).	Bendazoles.
<i>Trichuris trichiura</i> (whipworm)	Often asymptomatic; loose stools, anemia, rectal prolapse in children.	Fecal-oral.	Bendazoles.
Tissue			
<i>Toxocara canis</i>	Visceral larva migrans—migration into blood → inflammation of liver, eyes (visual impairment), CNS (seizures, coma), heart (myocarditis). Patients often asymptomatic.	Fecal-oral.	Bendazoles.
<i>Onchocerca volvulus</i>	Black skin nodules, river blindness (“black sight”).	Female black fly.	Ivermectin (ivermectin for river blindness).
<i>Loa loa</i>	Swelling in skin, worm in conjunctiva.	Deer fly, horse fly, mango fly.	Diethylcarbamazine.
<i>Wuchereria bancrofti</i> , <i>Brugia malayi</i>	Lymphatic filariasis (elephantiasis)—worms invade lymph nodes → inflammation → lymphedema E ; symptom onset after 9 mo–1 yr.	Female mosquito.	Diethylcarbamazine.



Cestodes (tapeworms)

ORGANISM	DISEASE	TRANSMISSION	TREATMENT
<i>Taenia solium</i> A	Intestinal tapeworm	Ingestion of larvae encysted in undercooked pork	Praziquantel
	Cysticercosis, neurocysticercosis (cystic CNS lesions, seizures) B	Ingestion of eggs in food contaminated with human feces	Praziquantel; albendazole for neurocysticercosis
<i>Diphyllobothrium latum</i>	Vitamin B ₁₂ deficiency (“Diphyllo B ₁₂ latum”; tapeworm competes for B ₁₂ in intestine) → megaloblastic anemia	Ingestion of larvae in raw freshwater fish	Praziquantel, niclosamide
<i>Echinococcus granulosus</i> C	Hydatid cysts D (“eggshell calcification”) most commonly in liver E and lungs; cyst rupture can cause anaphylaxis	Ingestion of eggs in food contaminated with dog feces Sheep are an intermediate host	Albendazole; surgery for complicated cysts

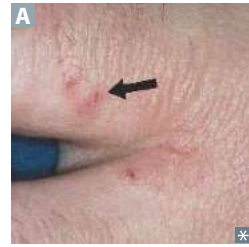
**Trematodes (flukes)**

ORGANISM	DISEASE	TRANSMISSION	TREATMENT
<i>Schistosoma</i>	Liver and spleen enlargement (A shows <i>S mansoni</i> egg with lateral spine), fibrosis, inflammation, portal hypertension; <i>S mansoni</i> and <i>S japonicum</i> can both also cause intestinal schistosomiasis, presenting with diarrhea, abdominal pain, iron deficiency anemia Chronic infection with <i>S haematobium</i> (egg with terminal spine B) can lead to squamous cell carcinoma of the bladder (painless hematuria) and pulmonary hypertension	Snails are intermediate host; cercariae penetrate skin of humans in contact with contaminated fresh water (eg, swimming or bathing)	Praziquantel
<i>Clonorchis sinensis</i>	Biliary tract inflammation → pigmented gallstones Associated with cholangiocarcinoma	Undercooked fish	Praziquantel



Ectoparasites

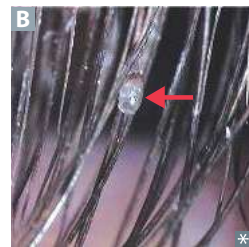
Sarcoptes scabiei



Mites burrow into stratum corneum and cause **scabies**—pruritus (worse at night) and serpiginous burrows (lines) often between fingers and toes **A**.

Common in children, crowded populations (jails, nursing homes); transmission through skin-to-skin contact (most common) or via fomites.
Treatment: permethrin cream, oral ivermectin, washing/drying all clothing/bedding, treat close contacts.

Pediculus humanus
and *Phthirus pubis*



Blood-sucking lice that cause intense pruritus with associated excoriations, commonly on scalp and neck (head lice), waistband and axilla (body lice), or pubic and perianal regions (pubic lice).

Body lice can transmit *Rickettsia prowazekii* (epidemic typhus), *Borrelia recurrentis* (relapsing fever), *Bartonella quintana* (trench fever).
Treatment: pyrethroids, malathion, or ivermectin lotion, and nit **B** combing. Children with head lice can be treated at home without interrupting school attendance.

Cimex lectularius and
Cimex hemipterus

Bed bugs. Blood-feeding insects that infest dwellings. Painless bites result in a range of skin reactions, typically pruritic, erythematous papules with central hemorrhagic punctum. A clustered or linear pattern of bites seen upon awakening is suggestive. Diagnosis is confirmed by direct identification of bed bugs in patient's dwelling.

Bed bugs can spread among rooms; cohabitants may exhibit similar symptoms. Infestations can also spread via travelers from infested hotels and the use of unwashed, used bedding.
Treatment: bites self resolve within 1 week. Eradication of the infestation is critical.

Parasite hints

ASSOCIATIONS	ORGANISM
Biliary tract disease, cholangiocarcinoma	<i>Clonorchis sinensis</i>
Brain cysts, seizures	<i>Taenia solium</i> (neurocysticercosis)
Hematuria, squamous cell bladder cancer	<i>Schistosoma haematobium</i>
Liver (hydatid) cysts, exposure to infected dogs	<i>Echinococcus granulosus</i>
Iron deficiency anemia	<i>Ancylostoma</i> , <i>Necator</i>
Myalgias, periorbital edema	<i>Trichinella spiralis</i>
Nocturnal perianal pruritus	<i>Enterobius</i>
Portal hypertension	<i>Schistosoma mansoni</i> , <i>Schistosoma japonicum</i>
Vitamin B ₁₂ deficiency	<i>Diphyllobothrium latum</i>